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Acta Tropica 97 (2006) 50–54
ACTA TROPICA
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Biting behavior and *Plasmodium* infection rates of *Anopheles arabiensis* from Sille, Ethiopia

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Received 30 November 2004; received in revised form 30 June 2005; accepted 2 August 2005
Available online 19 September 2005

Abstract

The man-biting behavior and *Plasmodium* infection rates of anopheline mosquitoes were investigated in Sille, a hyperendemic malarious area in southern Ethiopia. Seven *Anopheles* species were identified from all night landing collections, conducted from 18:00 to 06:00 h between October 2001 and August 2002. The predominant species was *Anopheles arabiensis* (55.8%), followed by *Anopheles coustani* (31.5%), *Anopheles pharoensis* (9.5%), *Anopheles funestus* (2.2%), *Anopheles nili* (0.5%), *Anopheles marshallii* (0.4%) and *Anopheles demeilloni* (0.2%). Dissection of *A. arabiensis* showed an average parous rate of 73.2%. A large proportion of the parous mosquitoes were caught biting in the latter part of the night. Malaria sporozoite rates were determined by ELISA for *A. arabiensis*, with 0.5% (4/796) infective with *Plasmodium falciparum* and 1.76% (14/796) with *Plasmodium vivax*; there were no mixed infections. From our small sample of sporozoite positives we found no association between biting behavior and sporozoite infection status.
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Keywords: Malaria; *Anopheles*; Parity; Mosquito; Sporozoite; Ethiopia

1. Introduction

Malaria is a disease of significant public health importance in Ethiopia. Salivary gland sporozoite rates and man–vector contact have shown *Anopheles arabiensis* as the major vector of malaria in the country, with *A. funestus*, *A. pharoensis* and *A. nili* playing secondary roles (Krafsur, 1971; Tulu, 1993; White et al., 1980). Determining the biting behavior, parous rate, sporozoite rate and behavior of infected and uninfected mosquitoes is important information for assessing the role of vectors in the epidemiology of malaria in the area. Such information should also assist in the development and implementation of local malaria control programs.

Few data are available on the age composition of *Anopheles* mosquitoes in Ethiopia. Parous rates ranging from 65–69% for *A. gambiae* s.l. and 34–43% for *A. pharoensis* were reported from the Awash area of southeastern Ethiopia (Rishikesh, 1966). Ameneshewa and Service (1996) reported an overall parous rate of 58.3% for *A. arabiensis* at Gergedi in the Awash valley. Abose et al. (1998) reported parous rates of 43.2 and 36.5% for *A. arabiensis* and *A. pharoensis*, respectively, from the

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0001-706X/$ – see front matter © 2005 Elsevier B.V. All rights reserved.
doi:10.1016/j.actatropica.2005.08.002

Zwai area. However, records regarding the biting behavior of the parous and malaria-infected mosquitoes are lacking.

Several investigations on the vectors of malaria in Africa and elsewhere have shown pronounced difference in the biting time of parous and nulliparous mosquitoes. Differences in the biting time of sporozoite-infected and uninfected mosquitoes have also been reported. Charlwood et al. (1986) reported a different biting cycle for parous and nulliparous *A. farauti* in Papua New Guinea. Compared to the nulliparous, an increased proportion of parous mosquitoes were collected later in the night. All night man-biting collections of *A. gambiae* and *A. punctulatus* in Sierra Leone and Papua New Guinea, respectively, have shown parous females with a tendency to bite later than the nulliparous (Bockarie et al., 1996). Investigators have also reported differences in the biting behavior of sporozoite infective and uninfective mosquitoes. A significantly declining trend in the sporozoite rates of *A. gambiae* during the night was observed by Gilles and Wilkes (1963) in Tanzania. On the contrary, an increasing sporozoite rates during the night were reported for *A. gambiae* and *A. funestus* in Burkina Faso (Robert and Carnevale, 1991) and for *A. punctulatus* in Papua New Guinea (Bockarie et al., 1996). Maxwell et al. (1998) from Tanzania on the other hand found no significant differences in sporozoite rates in different parts of the night. The present study was designed to investigate the feeding behavior of parous and nulliparous *Anopheles* species in Sille, Ethiopia, with an attempt to determine the biting behavior of sporozoite infective and uninfective mosquitoes.

2. Materials and methods

2.1. Study area

This study was conducted in Sille (5°53′N, 37°29′E; altitude of 1100 m), an agricultural area in southern Ethiopia approximately 525 km from Addis Ababa. This is a hyperendemic malarious area (Adugna et al., 1998). The local population consists primarily of farmers, daily/permanent laborers in the nearby state farm and fishermen. Houses are constructed of mud and wood, with grass thatch roofs. The swampy nature of the area combined with local irrigation contributes to high mosquito densities throughout most of the year.

2.2. Collection and identification of mosquitoes

Indoor and outdoor human landing collections of mosquitoes were made every two months, on four consecutive nights, from October 2001 to August 2002. Collections were made using aspirators and flashlights and mosquitoes were caught as they attempted to bite the collectors (entomology technicians from the local malaria control sector) and to avoid risk of malaria infection the collectors were provided with malaria prophylaxis. Catches were made over a continuous 12-h period, 18:00–06:00 h, with a change in collectors every 4 h. The hourly collections of anophelines were kept in separate labeled paper cups. Mosquitoes were provided with a sugar solution until identification and dissected for parity. Identification to the species level was made using keys of Verrone (1962) and Gilles and Coetzee (1987).

2.3. Parity determination and ELISA processing

All mosquitoes belonging to the *A. gambiae* complex were dissected for parity determination as described by Gilles and Wilkes (1963) and the ovaries were examined to establish parity according to the method of Detinova (1962). Hourly catches of anopheline mosquitoes were identified as either parous or nulliparous depending on whether the ovarian trachioles were coiled or uncoiled. Parous mosquitoes were stored over silica-gel dessicant until tested for circumsporozoite protein (CSP). Enzyme-linked immunosorbent assays (ELISA) were used to test heads and thoraces of the dried mosquitoes for the presence of *Plasmodium falciparum*, *P. vivax*-210 and *P. vivax*-247 CSP (Wirtz et al., 1987, 1992).

2.4. Statistical analysis

A 2×2 chi-square test was used for statistical analysis. All graphs were plotted using Microsoft Excel.

3. Results

A total of 1948 *Anopheles* mosquitoes belonging to seven species were collected: *A. gambiae* s.l. (assumed to be *A. arabiensis* from previous studies in the area), *A. pharoensis*, *A. funestus*, *A. nili*, *A. coustani*, *A. marshallii* and *A. demeilloni* (Table 1). The major malaria vector in the country, *A. arabiensis*, was the most abundant species collected accounting for 55.8% of the total, followed by *A. coustani* (31.5%) and *A. pharoensis* (9.5%). *A. funestus*, *A. nili*, *A. marshallii* and *A. demeilloni* were present in small numbers. The human biting rate for *A. arabiensis* varied from 19.3 bites/person/night in August to 82 bites/person/night in April. The average human

Table 1
Man-biting *Anopheles* species caught in Sille, Southern Ethiopia (October 2001–August 2002)

Date	Numbers of each species of *Anopheles*							
	A. arabiensis	*A. pharoensis*	*A. funestus*	*A. nili*	*A. coustani*	*A. marshallii*	*A. demeilloni*	Total[a]
October 2001	215	83	22	7	164	7	–	498 (25.6)
December 2001	89	26	4	–	145	–	1	265 (13.6)
February 2002	84	5	–	–	42	–	1	132 (6.8)
April 2002	329	20	5	–	28	–	–	382 (19.6)
June 2002	293	36	9	1	146	–	1	486 (24.9)
August 2002	77	15	3	1	89	–	–	185 (9.5)
Total	1087 (55.8)	185 (9.5)	43 (2.2)	9 (0.5)	614 (31.5)	7 (0.4)	3 (0.2)	1948 (100)

[a] Months catch as percentage of total collection.

biting rate for this species during the study period was 45.3 bites/person/night.

Table 2 shows the relative abundance of indoor and outdoor human bait collections of the anopheline mosquitoes at Sille. A preponderance of *A. arabiensis* was observed in the indoor collections. Among the total of 1087 *A. arabiensis*, more than half (589) were caught biting indoors, accounting for 81% of the total indoor anopheline catches. *A. coustani* was the commonest species in outdoor collections and 91% of the catch of this species was made outdoors indicating the strongly exophagic behavior of this species. Though small numbers were collected, both *A. pharoensis* and *A. funestus* were also observed to exhibit a tendency to exophagic behavior. Considerable differences were observed in the nocturnal biting cycles of *A. arabiensis*, *A. pharoensis* and *A. coustani* (Fig. 1). *A. arabiensis* bite during the entire night, with the peak occurring between 23:00 and 03:00 h. The *A. coustani* and *A. pharoensis* biting peak occurred between 18:00 and 20:00 h and decreased later but with a suggestion of an increase in the biting activity of *A. coustani* towards dawn. The biting cycle in the indoor and outdoor environment appeared to be similar.

Table 2
Relative abundance of indoor and outdoor human bait collections of anopheline mosquitoes at Sille (October 2001–August 2002)

Species	Number collected (each species as percentage of total collection)		
	Indoor	Outdoor	Total
A. arabiensis	589 (81)	498 (40.8)	1087 (55.8)
A. pharoensis	66 (9.1)	119 (9.7)	185 (9.5)
A. funestus	9 (1.2)	34 (2.8)	43 (2.2)
A. nili	6 (0.8)	3 (0.2)	9 (0.5)
A. coustani	55 (7.6)	559 (45.8)	614 (31.5)
A. marshallii	2 (0.3)	5 (0.4)	7 (0.4)
A. demeilloni	–	3 (0.2)	3 (0.2)
Total (percentage of total indoors and outdoors)	727 (37.3)	1221 (62.7)	1948 (100)

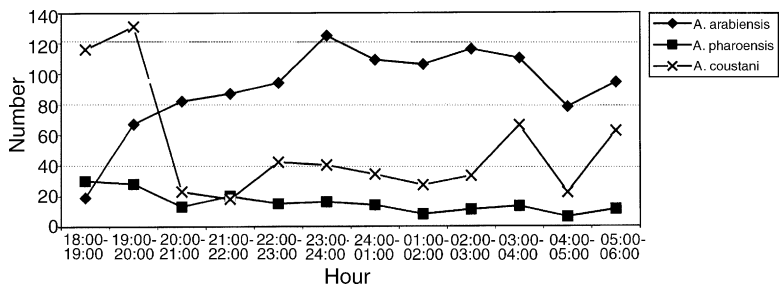


Fig. 1. The nocturnal biting cycle of *Anopheles arabiensis*, *A. pharoensis* and *A. coustani*.

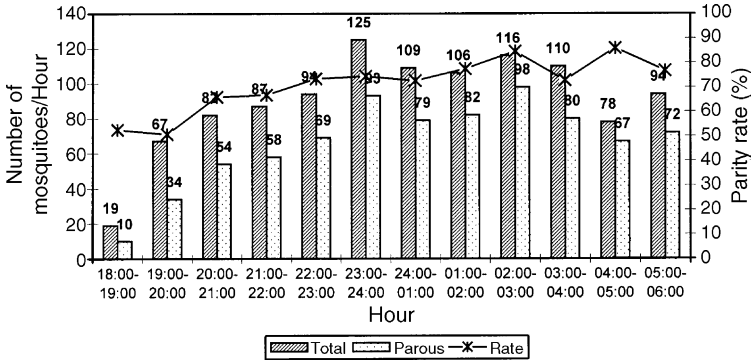


Fig. 2. Number and parity rates (%) of hourly collections of *Anopheles arabiensis*, from human landing collections made at Sille, Southern Ethiopia, October 2001–August 2002.

Table 3
Number and percentage of parous *Anopheles arabiensis* mosquitoes grouped into 4-h periods

	Collection time (h)			Total	χ^2 for trend	*P*-value
	18:00–22:00	22:00–02:00	02:00–06:00			
Mosquitoes collected	255	434	398	1087		
Percent of collection	23.5	39.9	36.6	100		
Percent parous	61.2	74.4	79.6	73.2	25.4	<0.0001
Mosquitoes assayed	156	323	317	796		
Sporozoite rate	1.92 (3/156)	2.78 (9/323)	1.89 (6/317)	2.26 (18/796)	0.042	0.84
95% CI	0.39–5.38	1.15–5.21	0.53–3.88	1.34–3.54		

Only *A. arabiensis* ($n = 1086$) were dissected for parity determination. The parity rate appeared to increase during the night (Fig. 2). When grouped into 4-h periods (Table 3), parity rates during the first, second and third groups were 61.2, 74.4 and 79.6%, respectively, suggesting an increase in the parity rate during the night (Mantel Haenszel χ^2 analysis for linear trend = 25.4, $P < 0.0001$).

Sporozoite ELISAs were conducted on 796 *A. arabiensis* and 18 (2.26%) mosquitoes were found infected (95% CI 1.34–3.54%). Of these eighteen, 14 were *P. vivax* and 4 were *P. falciparum*. No trend was observed in the biting time of sporozoite-infected and uninfected ones ($\chi^2 = 0.42$, $P = 0.84$).

4. Discussion

More than 55% of the man-biting anophelines collected in the present study belong to the *A. gambiae* complex. Previous investigations have shown *A. arabiensis* to be the only member of the *A. gambiae* complex in the study area (Lulu et al., 1999; Habtewold et al., 2001). Most biting by this species (76.5%) occurred after 22:00 h when people are presumed to be in bed. The second most abundant anopheline was *A. coustani*, which was caught in larger numbers early in the evening and also outdoors. This is similar to what was reported by Krafsur (1977), where only 17% of the human biting captures of *A. coustani* were indoors. A similar early evening outdoor biting trend for *A. coustani* was reported in Mozambique (Mendis et al., 2000). This species has not been incriminated as a vector of malaria in Ethiopia. However, the role that it might play in transmission needs further investigation, particularly taking into consideration the high outdoor biting density in the early part of the night that coincides with the outdoor activity of the local population. The over all mean parity rate of 73.2% for *A. arabiensis* in the study area indicates that the population is long lived and has the capacity to be an efficient vector of malaria. The parous rate observed is higher than the 58.3% reported by Ameneshewa and Service (1996) and the 43.2% rate reported by Abose et al. (1998), but comparable to the 65–69% rate which Rishikesh (1966) reported for this species in other parts of the country. A significantly larger proportion of parous *A. arabiensis* were caught biting in the latter part of the night. Comparable late biting of parous mosquitoes of the *A. gambiae* complex was reported from Burkina

Faso (Robert and Carnevale, 1991) and Sierra Leone (Bockarie et al., 1996).

The CSP rate of 2.26% in the present study is greater than those previously recorded in other parts of Ethiopia (Krafsur, 1977; Nigatu et al., 1994). The *P. falciparum* sporozoite rate of 0.5% is comparable with the 0.76% reported by Nigatu et al. (1994) from Gambella in the west and lower than the 1.1% from Arbaminch, which is close to the present study area (Habtewold et al., 2001). Although a significantly larger proportion of parous *A. arabiensis* were caught biting in the latter part of the night we found no trend in the biting behavior of sporozoite infective mosquitoes in the different parts of the night. The absence of a significant trend in the biting behavior of sporozoite-infected mosquitoes could be due to the small number of sporozoite positive individuals, similar to that reported for *A. gambiae* and *A. funestus* from Tanzania (Maxwell et al., 1998).

With the local human population in the study area, usually staying in bed between 22:00 and 06:00 h, only 17% (3/18) of the infective bites would occur before they retire to bed. Thus, if intact and/or insecticide treated bed nets are used 83% of the infective bites would probably be prevented. Consequently, the late biting behavior of the more dangerous parous mosquitoes is a significant finding with respect to the increased use of insecticide treated bed nets for the reduction of malaria.

Acknowledgements

The authors would like to express their gratitude to the malaria and other vector borne diseases control team of the Gamo Gofa zone, SNNPR, Health Bureau, for the support they provided during the study. The sporozoite ELISA reagents were produced with World Health Organization Collaboration Center support.

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